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Sentiment-driven community detection in a network of perfume preferences

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Abstract

The rapid growth of online perfume retail platforms has generated vast amounts of user-generated content, including reviews, ratings, and feedback, which contain valuable insights into consumer preferences. However, effectively analyzing this information to uncover hidden structures and provide accurate recommendations remains a major challenge. The primary objective of this study is to apply community detection techniques to group similar perfumes liked by users into meaningful clusters, thereby supporting more accurate perfume recommendations. This study proposes a novel methodology that integrates sentiment analysis and community detection within a Perfume Co-Preference Network constructed from Persian perfume reviews. First, a transformer-based Persian language model was employed to classify user comments into positive and negative sentiments with high accuracy. To enhance classification performance, emojis were mapped to their Persian semantic equivalents, capturing subtle emotional cues often overlooked in text-based analysis. In addition, user ratings for scent, longevity, and sillage were incorporated to refine sentiment classification results and improve network edge weighting. Based on the positive reviews, a bipartite user–perfume graph was constructed and projected into the Perfume Co-Preference Network, where perfumes are connected if they receive positive feedback from the same users. Community detection algorithms, including Louvain, Fastgreedy, Walktrap, and Spin-Glass were then applied to identify clusters of similar perfumes. The results show that the Spin-Glass algorithm achieved the highest modularity score, producing semantically meaningful clusters that aligned with established fragrance families. Furthermore, the inclusion of sentiment, emoji-based refinement, and rating integration significantly improved network modularity compared to baseline models. As an illustrative example to demonstrate interpretability beyond modularity, we included a section analyzing the 21 communities detected by Louvain, showing that they also correspond to attributes such as perfume gender, fragrance groups, and olfactory nature (e.g., warm, fresh). This research represents the first application of community detection techniques in the analysis of perfume networks and contributes a unique, sentiment-annotated dataset of Persian perfume reviews for future network science research. The proposed approach offers practical implications for personalized perfume recommendation, targeted marketing, and consumer behavior analysis in the fragrance industry.

Keywords Community detection, Perfume networks, Network analysis, Sentiment analysis, Co-preference network, Consumer behavior, Recommendation systems, Fragrance industry, Perfume context

Introduction

Network science has emerged as a powerful framework for analyzing complex systems across diverse domains, including social interaction, biology, economics, and technology (Su et al. 2014)–(Mao et al. 2017). A central challenge in this field is the identification of communities, defined as groups of nodes that are more densely connected internally than with the rest of the network (Girvan and Newman 2002; Fortunato 2010). Community detection enables the discovery of hidden mesoscale structures and has proven valuable in applications ranging from social platforms and biological pathways to recommender systems (Gao et al. 2022; Luo et al. 2014).

Perfumes represent a distinctive case for recommender systems because they are experiential goods whose appeal depends on highly subjective and emotional evaluations. Unlike books or movies, where content features can be extracted directly, perfumes lack easily measurable descriptors. Several perfume recommender systems have attempted to cluster fragrances based on their aromatic notes (Elabonga 2023)–(Tahir 2023). However, this strategy has limitations: the same note can be reproduced with different chemical compounds, yielding scents that differ significantly in perception. Moreover, specialists caution against relying solely on notes when selecting a fragrance, as consumer preferences often reflect emotions, memories, and identity expressions rather than compositional similarity (Lutan 2025). Thus, effective perfume recommendation must account not only for functional similarity but also for the affective and contextual dimensions that strongly influence consumer choice.

Our study addresses this challenge by introducing community detection on a perfume co-preference network constructed from user interactions and enriched with sentiment analysis. Nodes represent perfumes, and edges capture co-preferences derived from shared user engagement. By incorporating sentiment extracted from textual reviews, emojis, and rating-based refinements, the resulting sentiment-weighted networks capture both structural ties and the affective tone of evaluations (Mazza et al. 2023)–(Salehi et al. 2018). This integration enables recommender systems to move beyond feature matching toward guided discovery, preference refinement, and a more human-centered appreciation of perfumes. For practitioners, this approach provides a pathway to more accurate personalized recommendations, market segmentation, and trend analysis (Luo et al. 2014; Mazza et al. 2023; Zhou et al. 2012). For network science, it illustrates how affective information can shape mesoscale structures in consumer product networks.

This methodological novelty also provides insight into an important perspective for network science. Sentiment may act as a structuring force in co-preference networks, with perfumes clustering not only by shared users but also by the affective tone of evaluations. Although not directly tested as a mechanism here, this perspective highlights a promising direction for future research that bridges affective computing and network analysis.

To carry out this study, we collected a comprehensive dataset from a large Persian fragrance platform. The dataset includes user comments, ratings, and identifiers such as usernames and user IDs. Rich metadata about perfumes was also gathered, including

fragrance group, perfumer, brand name, country of origin, and release year. In addition, we recorded community-level signals such as vote polls for Scent, Longevity, Sillage, and Bottle Design, as well as contextual details on whether comments were standalone or replies. Sentiment analysis was applied to classify user evaluations and refine edge weights, creating a unique sentiment-annotated network resource.

Community detection algorithms including Louvain (Blondel et al. 2008), FastGreedy (Clauset et al. 2004), Walktrap (Pons and Latapy 2005), and Spinglass (Reichardt and Bornholdt 2006) were applied to these networks. To assess the quality of the detected communities, we complemented modularity evaluation with interpretability analysis. In particular, we examined the 21 communities identified by Louvain, comparing them to fragrance family, perfume nature, and gender orientation. This interpretive assessment confirmed that the detected communities correspond closely to consumer-facing categories and established fragrance taxonomies, providing external validation beyond modularity scores.

The contributions of this paper are fourfold:

- *Application novelty* We present the first application of community detection to perfume co-preference networks explicitly shaped by sentiment, addressing the limitations of purely note-based clustering by incorporating affective and experiential dimensions of fragrance perception.
- *Resource contribution* We construct and release a unique, sentiment-annotated dataset of Persian perfume reviews that combines user interactions, textual evaluations, emojis, and rating-based refinements. This resource provides a valuable foundation for future work in network science, recommender systems, and affective computing.
- *Sentiment as a structuring force* We demonstrate that affective signals, including emotions, memories, and perceptions expressed in reviews, can influence network organization and produce communities that align closely with fragrance families, gender orientation, and consumer-facing taxonomies.
- *Sentiment-driven refinement* We show that integrating sentiment analysis into network construction and edge weighting not only enhances modularity scores but also improves interpretability, enabling recommender systems to move beyond functional similarity toward guided discovery and individualized preference refinement.

The remainder of this paper is structured as follows: Sect. “[Related work](#)” reviews related work, Sect. “[Methodology](#)” outlines the methodology, Sect. “[Results](#)” presents the results and analysis, Sect. “[Conclusions and future work](#)” concludes with key findings, limitations, and directions for future research.

Related work

Community detection has long been a central problem in network science, providing insights into the intermediate-scale organization of complex systems and enabling the discovery of hidden structures across domains from sociology to biology and economics (Fortunato 2010; Fortunato and Hric 2016). A wide range of approaches has been developed, but methods based on modularity optimization and dynamic processes have proven especially effective for large-scale and heterogeneous networks. The Louvain

algorithm (Blondel et al. 2008) and the FastGreedy method (Clauset et al. 2004) exemplify modularity-based approaches, both designed to maximize the modularity score that compares observed edge densities to those expected in random graphs (Newman and Girvan 2004). Louvain is highly scalable, making it suitable for large, weighted networks, while FastGreedy is efficient for sparse or hierarchical structures. Dynamic methods offer a complementary perspective by modeling processes such as random walks or spin dynamics on the network. The Walktrap algorithm (Pons and Latapy 2005) exploits the tendency of short random walks to remain within the same community, thereby capturing local structural coherence, while Spinglass methods (Reichardt and Bornholdt 2006; Wu 1982) frame community detection as an energy minimization problem with a resolution parameter that enables the exploration of communities at multiple scales. Together, modularity-based and dynamic approaches balance scalability and interpretability, making them well suited for preference networks where subtle but meaningful partitions are expected.

Within recommender systems, community detection has been shown to improve personalization by identifying groups of users or items with shared interaction patterns. Gasparetti et al. (Gasparetti et al. 2021) provided one of the earliest surveys of community detection in social recommender systems, emphasizing that modeling user–user and user–item interactions through communities can enhance accuracy and diversity in recommendations. Yang et al. (Yang et al. 2011) advanced this perspective by applying Bayesian probabilistic models to detect latent communities in social networks, showing that incorporating hidden preference structures leads to measurable gains in predictive performance. These studies highlight how network-based community modeling can bridge explicit behavior with underlying preferences, a principle that directly motivates applications in domains where user evaluations are subjective and context-dependent.

Perfume recommendation exemplifies such a domain. Perfumes are experiential goods, shaped not only by olfactory notes but also by personal memory, emotion, and cultural context. Traditional recommender approaches have clustered perfumes by aromatic notes (Elabonga 2023)–(Tahir 2023) yet experts caution that similar notes can be composed from distinct chemical compounds that evoke very different scents. Consequently, note-based similarity alone often fails to capture the true drivers of consumer choice (Lutan 2025). Research on product networks offers useful parallels: Gao (2023), for instance, analyzed co-purchasing networks on e-commerce platforms and showed how product communities emerge from collective consumer behavior rather than intrinsic product attributes. This insight can be adapted to perfumes by modeling how users evaluate and co-prefer fragrances, shifting attention away from chemical composition toward interaction-driven structures. Recent recommender studies have extended this logic further by incorporating perception-centered signals, such as mood, personality, and emotion (Pang and Lee 2008; Park 2020), highlighting the importance of affective dimensions in experiential product domains.

Sentiment analysis provides a natural entry point for capturing such dimensions. Foundational work by Pang and Lee (2008) demonstrated how textual sentiment classification can extract evaluative signals from product reviews, while Cambria (2016) broadened this field through affective computing approaches that account for emotional context. Wang et al. (2017) further illustrated how integrating sentiment with community detection allows for the profiling of communities not only by structural similarity but also by

collective emotional tone, improving the interpretability of recommendation models. In perfume networks, this integration is particularly valuable because user reviews often combine descriptive features with affective reactions, including the use of emojis as non-textual emotional markers. By leveraging sentiment-rich signals, networks can capture both structural co-preference and affective alignment. Our work builds on this direction by classifying Persian-language perfume reviews with ParsBERT (Farahani et al. 2021), a transformer model (Vaswani et al. 2017) tailored to the linguistic properties of Persian. Unlike multilingual models that often struggle with non-Latin scripts, ParsBERT provides accurate representations of nuanced morphology and syntax, making it particularly well suited for sentiment classification in this context. Combined with emoji mapping and rating-based refinements, this enables the construction of sentiment-weighted perfume co-preference networks that reflect how perfumes cluster not only through shared users but also through the affective signals embedded in their evaluations.

Methodology

To ensure clarity and ease of navigation, the methodology is organized into five subsections: A. Data Collection, B. Data Preprocessing, C. Sentiment Analysis, D. Network Construction, and E. Community Detection. Each subsection details a distinct stage of the pipeline, while Figs. 7 and 8 provide schematic overviews that visually summarize the process. Figure 7, placed at the end of Section C, illustrates the workflow from web scraping through sentiment classification, and Fig. 8, placed at the end of Section E, depicts the steps from network construction to community detection.

Data collection

Data collection was conducted from the Atrafshan website, a popular Iranian platform for perfume sales, reviews, and ratings (Atrafshan 2024). In total, 36,434 comments were extracted from 7387 unique users on 1369 unique perfumes, focusing on the most viewed products (Fig. 1).

For each perfume, its dedicated page (Fig. 2) was analyzed to locate the comment section and capture both user-level and perfume-level information. From user comments we recorded usernames, user IDs, review text, and individual ratings on four categories: Scent, Longevity, Sillage (the scent trail left by a perfume), and Bottle Design. A typical user review contained these detailed scores (Fig. 3). At the perfume level, we collected metadata such as fragrance group, perfumer, brand name (in Persian), country of origin, and release year, as well as overall voting polls summarizing average community ratings across the four categories. For example, the aggregated ratings for Lancome *La Vie Est Belle* are shown in Fig. 4.

We also noted whether each comment was a reply or a standalone review, ensuring discussion context was preserved. To guarantee integrity and completeness, duplicate entries and incomplete records were excluded, and error-handling measures were incorporated during collection. The final dataset combines detailed user evaluations with rich perfume metadata, providing a foundation for constructing sentiment-weighted co-preference networks. To support further research, the dataset has been shared in our public GitHub repository (Kalashi et al. 2025).

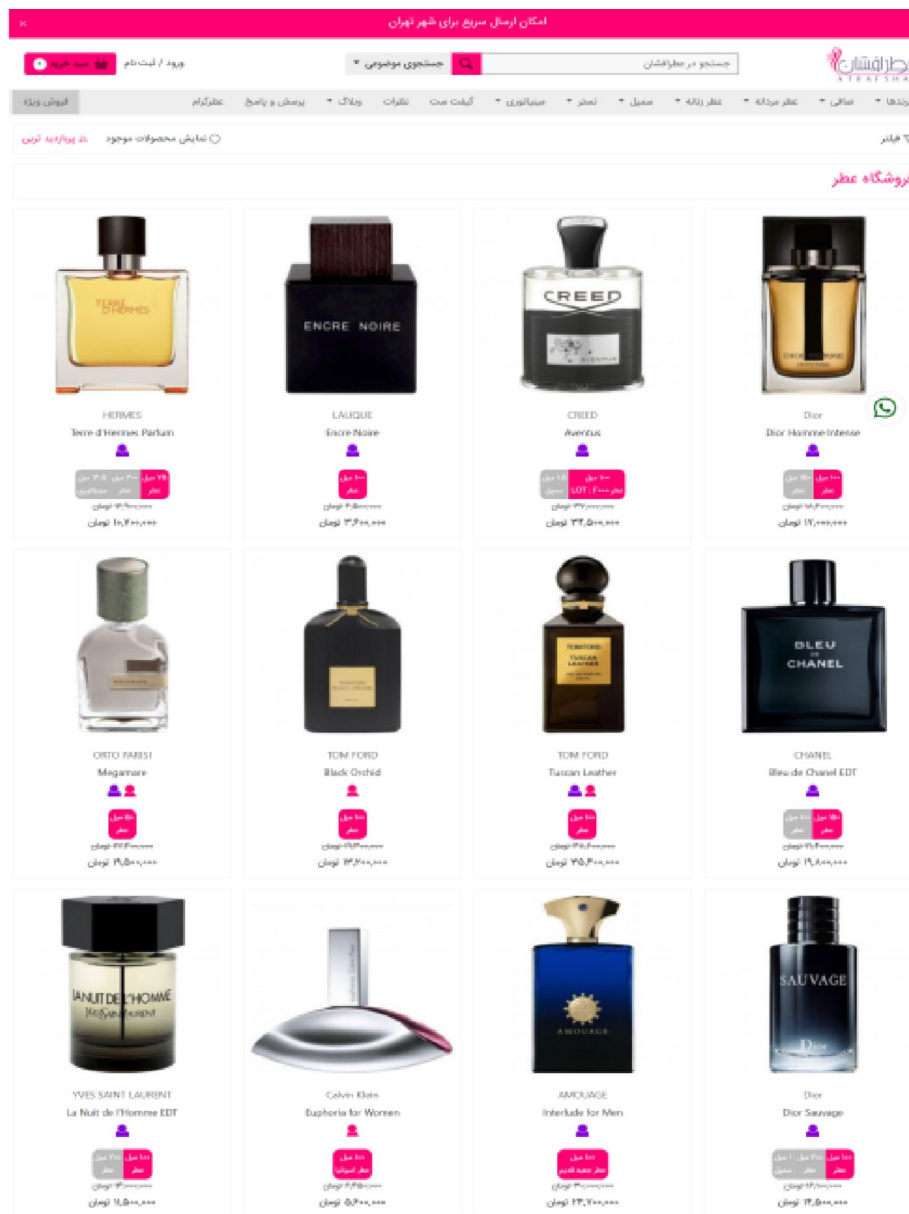


Fig. 1 First page of most viewed perfumes of Atrafshan website

Data preprocessing

Preprocessing was performed to ensure that the dataset was consistent, accurate, and suitable for network construction. Three main steps were applied. First, to avoid overlap between numerical user IDs and perfume IDs, unique prefixes were introduced. User IDs were reformatted as “user_x” and perfume IDs as “perfume_x,” ensuring they were treated as distinct entities within the bipartite network. For example, the user ID 4225 became user_4225, while the perfume ID 4225 became perfume_4225. This step eliminated potential ambiguity and was validated to confirm no duplication remained.

Second, missing and duplicate entries were addressed. Rows with absent User ID or Perfume ID were removed, while repeated user–perfume interactions were eliminated to avoid inflating co-preference links. This cleaning ensured the reliability of the dataset and prevented distortions in the resulting network.

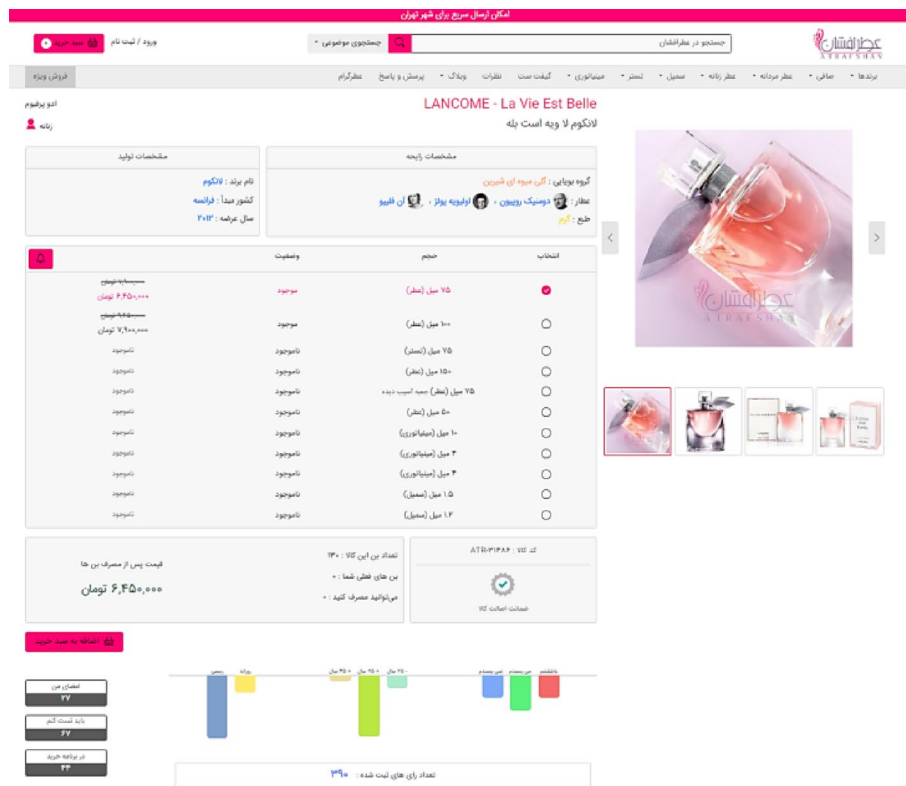


Fig. 2 Lancome La Vie Est Belle perfume page on Atrafshar

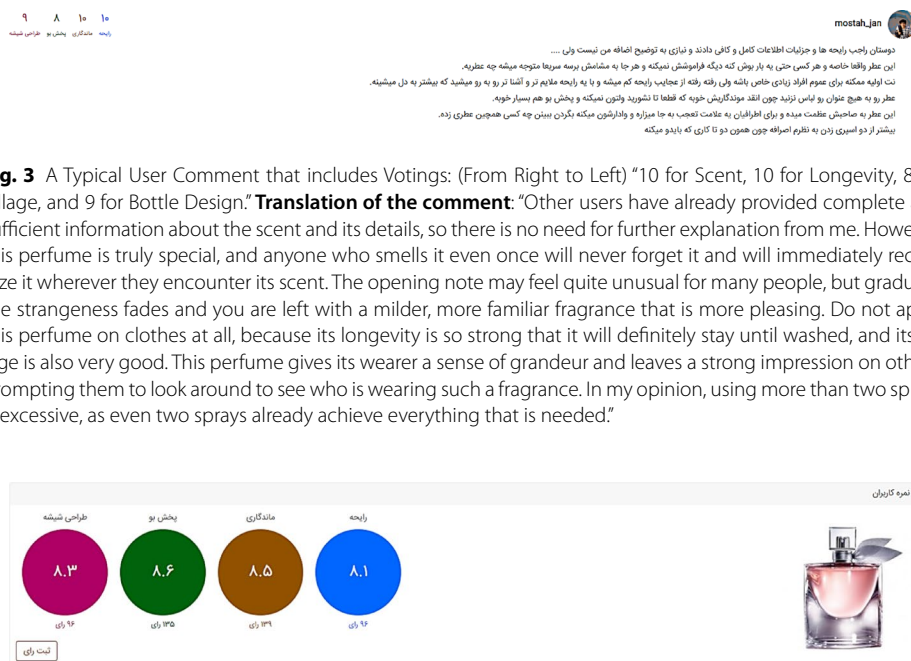


Fig. 3 A Typical User Comment that includes Votings: (From Right to Left) “10 for Scent, 10 for Longevity, 8 for Sillage, and 9 for Bottle Design.” **Translation of the comment:** “Other users have already provided complete and sufficient information about the scent and its details, so there is no need for further explanation from me. However, this perfume is truly special, and anyone who smells it even once will never forget it and will immediately recognize it wherever they encounter its scent. The opening note may feel quite unusual for many people, but gradually the strangeness fades and you are left with a milder, more familiar fragrance that is more pleasing. Do not apply this perfume on clothes at all, because its longevity is so strong that it will definitely stay until washed, and its sillage is also very good. This perfume gives its wearer a sense of grandeur and leaves a strong impression on others, prompting them to look around to see who is wearing such a fragrance. In my opinion, using more than two sprays is excessive, as even two sprays already achieve everything that is needed.”



Fig. 4 Average total user ratings for Lancome La Vie Est Belle perfume: (From Right to Left) “8.1 for Scent, 8.5 for Longevity, 8.6 for Sillage, and 8.3 for Bottle Design.”

Table 1 Statistics of the primary and filtered bipartite user-perfume network data

	Comments	Unique users	Unique perfumes
Primary dataset	36,434	7387	1369
Removing perfumes with 3 or fewer comments	35,910	7387	1104

Table 2 Example emoji mapping: selected emojis with corresponding Persian phrases and English translations (full emoji-to-Persian Dictionary is Available on our github repository, linked in the Data Collection section of our paper.)

EMOJI	CORRESPONDING PERSIAN PHRASE	ENGLISH TRANSLATION
👍	پسندیدن	LIKE
😡	عصبانی	ANGRY
❤️	عشق	LOVE
👏	عالی	GREAT
😞	ناامید	DISAPPOINTED
🤢	حالت تهوع	NAUSEATED

Finally, perfumes with very low engagement were excluded. Specifically, perfumes that had received three or fewer comments were filtered out, as such limited interactions were unlikely to provide meaningful information about user preferences. This step reduced noise and improved the robustness of the bipartite user–perfume network. The overall effect of these preprocessing steps is summarized in Table 1, which reports statistics for the dataset before and after filtering, highlighting the reduction of sparsity and the strengthening of user–perfume connections.

Sentiment analysis

Sentiment analysis was applied to user comments to capture their emotional tone and classify each review as positive or negative. The process consisted of three main steps: text preprocessing, sentiment classification with ParsBERT (Farahani et al. 2021), and integration of user voting to refine results. Text preprocessing was carried out using Hazm (Roshan Research 2024), a Persian language toolkit that standardizes text by correcting spacing errors, normalizing diacritics, and unifying numerals and punctuation. Since emojis were frequently used in reviews, we constructed a dictionary of 392 emojis mapped to equivalent Persian phrases, verified through Emojipedia (Emojipedia 2024). For example, the emoji “👁️” was translated to “چشم های عاشق” (“loving eyes”), ensuring that emotional context was preserved. Emojis were then replaced in comments with their Persian equivalents, joined with “,” (“and”), so that their affective meaning was retained. Table 2 illustrates examples of common emojis and their mappings, with the complete dictionary available in our GitHub repository (Kalashi et al. 2025).

Table 3 Statistics of sentiment classification results

		Positive	Negative
ParsBERT		28,757	7674
Integration of User Voting	Scent	28,341	8090
	Longevity	29,532	6899
	Sillage	29,393	7038



Fig. 5 Example of Sentiment Shift from Positive to Negative After User Vote Integration for both Longevity and Sillage (The user voted “1” for longevity and “3” for Sillage). The initial positive sentiment was influenced by the scent and initial sillage, but the user later expressed disappointment regarding its longevity. **Translation of the comment:** “I recently got this perfume. Its fragrance is nice, and the projection is good for the first half hour, but its longevity is only about 2 h.”

For sentiment classification, we used ParsBERT (ParsBERT 2024), a transformer model tailored for Persian text. Processed comments were tokenized and passed to a feedforward classifier that assigned binary labels of positive or negative. Batch processing was employed to handle the dataset efficiently, maintaining both speed and accuracy.

To refine these classifications, user ratings for Scent, Longevity, and Sillage were incorporated into the sentiment label S (the sentiment label assigned to a perfume aspect). For each rating R (the user rating assigned to a specific aspect of the perfume), the decision rules were applied consistently across categories and can be summarized as:

$$S(R) = \begin{cases} \text{Positive} & \text{if } R \geq 7 \\ \text{Negative} & \text{if } R \leq 3 \\ \text{Ambiguous} & \text{if } 4 < R < 7 \end{cases}$$

A rating of 7 or higher confirmed the sentiment as positive, while a rating of 3 or lower overrode the label as negative. Ratings between 4 and 6 were classified as ambiguous and left unchanged, ensuring that neutral or unclear cases did not distort the analysis. This systematic integration of structured votes with text-based sentiment allowed us to capture situations where a user’s written review and numeric ratings diverged.

Table 3 presents the overall sentiment classification statistics. Figures 5 and 6 show representative cases where integration of votes shifted the sentiment from positive to negative and vice versa. Figure 7 provides a schematic overview of the methodology, illustrating the process from web scraping to final sentiment classification. We also share the results of our sentiment analysis for each category with the community via our GitHub repository (Kalashi et al. 2025), which is also linked in the Declarations section of the paper.

Network construction

The analysis began with the construction of a bipartite user–perfume network, where one set of nodes represented users and the other represented perfumes. Edges were formed when a user left a positive comment on a perfume, capturing preference relations. Initially, the network contained 7387 unique users and 1104 perfumes, resulting in 8491 nodes and 35,910 edges. After filtering for positive sentiment, three separate bipartite networks were built corresponding to the categories Scent, Longevity, and Sillage. Table 4 reports statistics for these networks, which provided the basis for subsequent



Fig. 6 Example of Sentiment Shift from Negative to Positive After User Vote Integration for All Categories. (The user voted “9” for all three essential categories.) The ParsBERT model initially misclassified the comment due to its complexity and nuanced sentiment, as the user shared mixed emotions about the perfume’s initial and later impressions. **Translation of the comment:** “Hello, dear Atrafshan users. I recently purchased this perfume from the site. At first, I was a little disappointed because when I used it and smelled it up close, I didn’t like it; it had a different, familiar, and unpleasant scent up close. But I realized that I only smell this scent when I am very close, and from half a meter away the smell is completely different. It was like magic for me. I tried this perfume on someone else with one spray, and they had the same opinion, that it initially disappoints up close but then immediately starts to charm you. The more time passes, the more pleasant and attractive it becomes. In the first 2 to 3 h, it has a scent somewhat like “Midnight Rose”, although it is clear that they have their differences. However, with just one spray, it has far better projection and longevity compared to “Midnight Rose”. Part of my initial disappointment was due to its similarity to “Midnight Rose”, which felt repetitive to me, but in any case, it has become one of my favorite perfumes. With just one spray, it still has good, satisfying projection after 3 days. Finally, I thank the Atrafshan website.”

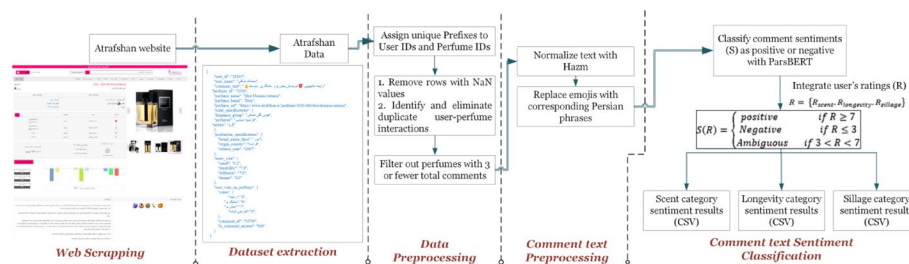


Fig. 7 Schematic representation of our methodology: from web scraping to comment text sentiment classification

Table 4 Statistics of the bipartite user-perfume networks and corresponding perfume co-preference networks

		Nodes		Edges	Weighted	Undi- rected
		User	Perfume			
Bipartite user-per- fume graph	Primary	7387	1369	36,434	NO	YES
	Removing perfumes with three or fewer comments	7387	1104	35,910	NO	YES
	Positive Scent	6191	1008	27,402	NO	YES
	Comments Longevity	6493	1015	28,600	NO	YES
Perfume Co-Preference (Without self-loop)	Sillage only	6477	1012	28,458	NO	YES
	Scent	–	1008	169,916	YES	YES
	Longevity	–	1015	172,884	YES	YES
	Sillage	–	1012	173,206	YES	YES

transformations. Figure 7 (from the previous section) schematically illustrates this pipeline.

The bipartite structure was then projected into a perfume co-preference network, where nodes represent perfumes and edges capture shared positive feedback. This projection was achieved using matrix multiplication. Let the bipartite matrix be denoted as A , Where n is the number of users, m is the number of perfumes and each element A_{ij} is 1 if user i has commented positively on perfume j , and 0 otherwise.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

First, the transpose of the user-perfume bipartite matrix, A^T , is calculated:

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{bmatrix}$$

Now, A^T has dimensions $m \times n$, where each row corresponds to a perfume and each column to a user. Next, matrix multiplication between A^T and the original matrix A is performed:

$$P = A^T \cdot A$$

This matrix multiplication results in an $m \times m$ perfume-perfume adjacency matrix P , Where:

$$P_{jk} = \sum_{i=1}^n A_{ji}^T \cdot A_{ik}$$

- P_{jk} is the number of users who reviewed both perfume j and perfume k .
- A_{ji}^T is the element in the j -th row and i -th column of A^T .
- A_{ik} is the element in the i -th row and k -th column of A .

Diagonal elements P_{jj} represent the total number of positive reviews for perfume j , but since these self-loops only reflect individual popularity rather than relational structure, they were removed before community detection. From the three bipartite networks, three corresponding co-preference networks were constructed, and their statistics are also shown in Table 4. These networks capture how perfumes cluster based on overlapping positive evaluations.

To refine edge weights and better reflect preference strength, category-specific average ratings were incorporated. To determine the final edge weight W_{jk} between two perfumes j and k , we calculate it using the following formula:

$$W_{jk} = (0.6 \times R_j + 0.6 \times R_k) + (0.4 \times C_{jk})$$

In this equation:

- R_j and R_k represent the category-specific average ratings (e.g., for Scent, Longevity, or Sillage) for perfumes j and k respectively.
- C_{jk} represents the original adjacency value, indicating the number of users who commented positively on both perfumes j and k .

This weighting scheme was selected heuristically after experimenting with several alternatives during development. We observed that giving greater emphasis to average ratings while retaining co-preference counts led to higher modularity and produced

communities that were more coherent and interpretable, often aligning with recognizable fragrance families and usage attributes. Although not statistically optimized, this balance provided a practical way to enhance the clarity and usefulness of the detected clusters.

Community detection

Community detection was applied to the three sentiment-specific co-preference networks, corresponding to Scent, Longevity, and Sillage. To ensure coverage of different methodological perspectives, we used two modularity-based algorithms, Louvain and Fast Greedy, and two dynamic approaches, Walktrap and Spinglass. For each case, modularity was computed to evaluate partition quality and confirm that the detected groups of perfumes reflected meaningful patterns of user preferences. As illustrated in Fig. 8, this process was carried out separately for each sentiment dimension, preserving the specificity of evaluations across categories. Before applying the algorithms, the networks were refined using the 60–40 weighting scheme described in Section D, followed by the removal of edges with weights of three or less and the elimination of isolated nodes. To balance the influence of highly connected perfumes, edge weights were further normalized by node degrees according to:

$$w_{\text{normalized}} = \frac{w_{\text{combined}}}{\text{degree}_u \times \text{degree}_v}$$

where degree_u and degree_v represent the degrees of the two connected perfumes. This adjustment reduced popularity bias while maintaining co-preference structure. The application of these algorithms produced community structures that offered interpretable and category-specific insights into perfume preferences, forming the basis for the analysis presented in later sections.

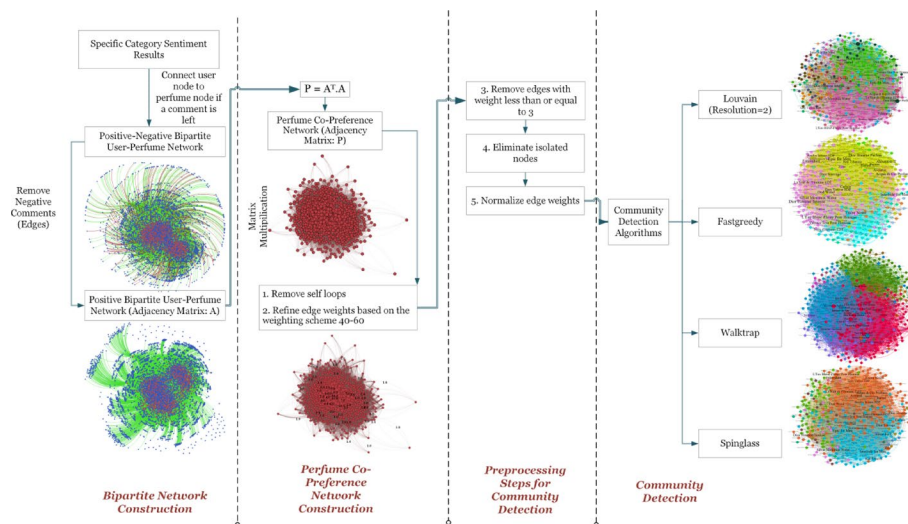


Fig. 8 Schematic representation of our methodology: from network construction to community detection. Note that this process is conducted individually for the sentiment results specific to each category

Results

Modularity trends across methods and networks

We first assessed the impact of sentiment analysis and edge-weight refinement on modularity. Table 5 reports the modularity scores across all algorithms and network types. The primary network, which included both positive and negative comments, achieved a modularity of 0.41 using the Louvain method, with an average of 0.43 across all algorithms. When sentiment analysis was incorporated and networks were reconstructed using only positive comments, modularity scores improved. Louvain scores increased to 0.44–0.46, representing a 7–12% gain, while the average modularity across methods rose to 0.47. This refinement was applied individually to the Scent, Longevity, and Sillage networks to capture dimension-specific user preferences.

Further improvement was observed when edge weights were enhanced with average user ratings. In this case, Louvain modularity reached 0.52–0.53, an overall increase of 27–29% compared to the primary network. The average modularity across all algorithms rose to 0.53, with values ranging from 0.48 for Walktrap to 0.58 for Spinglass. These findings show that combining sentiment analysis with rating-based edge refinement yields more cohesive and interpretable community structures, consistently outperforming the baseline networks.

Table 5 The statistics of community detections

Method	Network construction type		Modularity	Number of communities detected
LOUVAIN (Resolution = 2)	Primary		0.41	30
	Integration of User's comment Sentiment Analysis	Scent	0.45	30
		Longevity	0.46	34
		Sillage	0.44	34
	Integration of User's comment Sentiment Analysis + Perfume Total Voting in edge weights	Scent	0.53	21
		Longevity	0.52	24
		Sillage	0.52	24
FAST GREEDY	Primary		0.48	14
	Integration of User's comment Sentiment Analysis	Scent	0.51	18
		Longevity	0.52	16
		Sillage	0.51	15
	Integration of User's comment Sentiment Analysis + Perfume Total Voting in edge weights	Scent	0.54	21
		Longevity	0.55	19
		Sillage	0.56	17
WALK TRAP	Primary		0.44	3
	Integration of User's comment Sentiment Analysis	Scent	0.46	3
		Longevity	0.47	3
		Sillage	0.47	3
	Integration of User's comment Sentiment Analysis + Perfume Total Voting in edge weights	Scent	0.49	4
		Longevity	0.48	4
		Sillage	0.48	3
SPIN GLASS	Primary		0.49	8
	Integration of User's comment Sentiment Analysis	Scent	0.52	12
		Longevity	0.57	10
		Sillage	0.52	12
	Integration of User's comment Sentiment Analysis + Perfume Total Voting in edge weights	Scent	0.58	10
		Longevity	0.58	11
		Sillage	0.58	10

Performance of community detection algorithm

Comparisons across methods revealed that Spinglass achieved the highest modularity (up to 0.58), followed by Fast Greedy (0.56) and Louvain (0.53). Walktrap scored lower, with values near 0.48, reflecting its sensitivity to local structures. Figure 9 provides a visualization of the Louvain-detected communities in the Scent network, showing how sentiment-weighted and rating-refined edges contribute to clearer partitions. These results suggest that modularity-based methods (Louvain, Fast Greedy) and dynamic approaches (Spinglass) are best suited to sentiment-weighted co-preference networks.

Interpretability of detected communities

While modularity provides a useful quantitative benchmark, it does not guarantee that detected communities correspond to meaningful real-world structures. To evaluate interpretability, we analyzed the 21 communities detected by Louvain in the Scent network, focusing on three key perfume attributes: fragrance family (floral, oriental, chypre, woody, aromatic, leather, citrus), perfume nature (fresh, warm, moderate), and gender orientation (women, men, unisex). Table 6 summarizes the composition of each community, reporting the dominant category for each attribute and its percentage. To illustrate these distributions visually, three stacked column charts are provided: Fig. 10 shows the distribution of fragrance groups, Fig. 11 shows perfume nature, and Fig. 12 shows gender orientation across all communities.

The results demonstrate strong alignment between detected communities and established fragrance taxonomy. Community 0 consists of 14 perfumes, almost entirely women's (93%), with moderate floral styles such as *Miss Dior Blooming Bouquet*, *Mon*



Fig. 9 Visualization of communities detected by the Louvain method in the scent network, incorporating user comment sentiment analysis and perfume total voting in edge weights

Table 6 Summary of perfume communities detected by the Louvain method in the scent network, incorporating user comment sentiment analysis and perfume total voting in edge weights

Community	Population (perfumes)	Dominant Gender (%)	Dominant Nature (%)	Dominant Fragrance group (%)	Members
C0	14	Women (93%)	Moderate (50%)	Floral (43%)	'Gran Ballo', 'Lira', 'La Tosca', 'Olympéa', 'Bouquet Ideale', 'Narciso Rodriguez for Her EDP', 'Lost Cherry', 'Zen for Women', 'Narciso Poudree', 'Miss Dior Blooming Bouquet', 'Womanity EDP', 'My Way', 'Mon Guerlain', 'Burberry Her'
C1	2	Women (100%)	Fresh (100%)	Floral (100%)	'Chance Eau Tendre', 'Daisy'
C2	6	Men (83%)	Fresh (67%)	Oriental–Aromatic (Both 33%)	'Purpose', 'Dunhill Desire', 'Aventus', '212for Men', 'Dunhill Desire Blue', 'Aventus Cologne'
C3	27	Men (89%)	Warm (41%)	Woody (41%)	'Le Male Le Parfum', 'Bvlgari Man Wood Neroli', 'Versace Pour Homme Dylan Blue', 'L'Homme Ideal L'Intense', 'Eau Fraiche', 'Bitter Peach', 'Dunhill Icon', 'Wanted by Night', 'Uomo Salvatore Ferragamo Urban Feel', 'Versace Eros Pour Homme', 'The One for Men EDP', 'Le Male', 'Bleu de Chanel Parfum', 'Megamare', 'Light Blue Eau Intense Pour Homme', 'Explorer', 'Declaration d'Un Soir', 'Acqua di Giò Profondo', 'Green Irish Tweed', 'Gentleman Eau de Parfum Reserve Privée', 'Sauvage Elixir', 'CH Men Prive', 'Ombre Leather Parfum', 'Bentley For Men Black Edition', 'Y Eau de Parfum', 'Bleu de Chanel EDT', 'Y Le Parfum'
C4	7	Unisex (100%)	Warm (71%)	Oriental (43%)	'Enclave', 'Greenley', 'Jubilation for Men', 'The Library Collection Opus XIV Royal Tobacco', 'Interlude 53', 'Search', 'Tuxedo'

Table 6 (continued)

Community	Population (perfumes)	Dominant Gender (%)	Dominant Nature (%)	Dominant Fragrance group (%)	Members
C5	74	Unisex (62%)	Moderate (60%)	Oriental (42%)	'Viride', 'Dior Sauvage', 'Cuoium', 'Gucci Guilty Absolute', 'Journey for Men', 'Uomo Trussardi 2011', '02 L'Air du Desert Marocain', 'Arabian Desert', 'Beach Hut Man', 'Carlisle', 'Psychedelique', 'Mefisto Gentiluomo', 'Rosendo Mateu N° 5 Floral, Amber, Sensual Musk', 'Portrayal Man', 'Blamage', 'David Yurman Limited Edition', 'Au Coeur du Desert', 'Oud for Greatness', 'Nisean', 'Side Effect', 'Soleil de Jeddah', 'Reflection for Men', 'Club de Nuit Intense for men', 'The Tragedy of Lord George', 'Angels' Share', 'Honour for Men', 'Fiero', 'Imperial Millesime', 'Laudano Nero', 'Cedrat Boise', 'L'Homme Libre', 'Ambre Sultan', 'Epic for Men', 'Kirke', 'Boccanera', 'Akdeniz', 'Terroni', 'Meander', 'African Leather', 'Seminalis', 'Oajan', 'Italian Leather', 'Ganymede', 'Red Tobacco', 'Oud Essentiel', 'Hacivat', 'Memoir for men', 'XJ 1861 Renaissance', 'Percival', 'Tuscan Leather', 'Elise', 'Baronda', 'Oud Wood', 'Sauvage EDP', 'Baccarat Rouge 540', 'Layton Exclusif', 'XJ 1861 Naxos', 'Egoiste Platinum', 'Dignified', 'Haltane', 'Mefisto', 'Interlude Black Iris Man', 'Layton', '7 Anonimo', 'The Library Collection Opus VII', '1740 Marquis de Sade', 'Ombre Leather 2018', 'Epic for Women', 'Duro', 'Viking', 'Irish Leather', 'Baccarat Rouge 540 Extrait de Parfum', 'Herod', 'Oud Save The King'

Table 6 (continued)

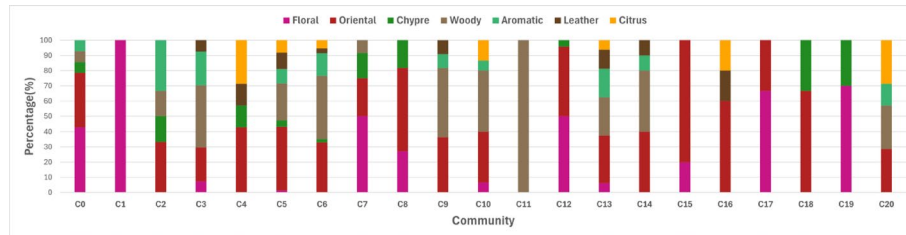
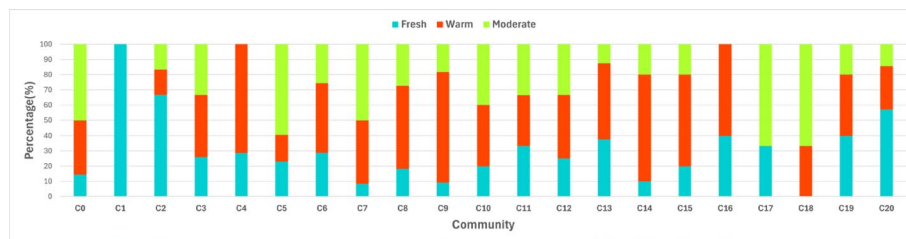
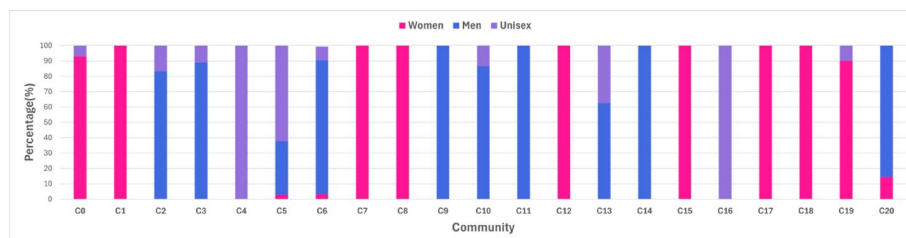
Community	Population (perfumes)	Dominant Gender (%)	Dominant Nature (%)	Dominant Fragrance group (%)	Members
C6	94	Men (87%)	Warm (46%)	Woody (42%)	'Santal Royal'; 'CH for Men'; 'Bentley Infinite Intense'; 'Jaguar for Men'; 'Allure Homme Sport'; 'Black XS for Men'; 'Yatagan'; 'The Library Collection Opus VIII'; 'Coco Noir'; 'Neroli Portofino'; 'Bijan for Men'; 'Guerlain Homme Intense EDP'; 'Tobacco Vanille'; 'Invictus'; 'Polo Blue'; 'Tom Ford Noir EDP'; 'Original Vetiver'; 'Silver Mountain Water'; '1 Million'; 'Allure Homme Sport Eau Extreme'; 'Oud Malaki'; 'La Nuit de l'Homme Le Parfum EDP'; 'Halloween Man'; 'A*Men'; 'CK One Shock for Men'; 'Lalique Pour Homme EDP'; 'Hugo Man'; 'Mine Pour Homme'; 'Givenchy pi'; 'Burberry London for Men'; 'Mercedes Benz Intense'; 'The Library Collection Opus V'; 'Acqua Di Gio Absolu'; 'Solo Platinum'; 'Only The Brave'; 'Shagya'; 'Bentley Absolute'; 'Encre Noire Sport'; 'Fahrenheit'; 'Jaguar Classic Gold'; 'Azzaro Pour Homme'; 'Uomo Salvatore Ferragamo'; 'Cool Water for Men'; 'He Wood Rocky Mountain Wood'; 'Happy for Men'; 'White Patchouli'; 'L'Eau d'Issey Pour Homme'; 'Aromatics Elixir EDP'; 'Bentley Intense EDP'; 'Spicebomb'; 'Fahrenheit Le Parfum'; 'Narciso Rodriguez for Him Bleu Noir'; 'Silver Scent'; 'Chrome'; 'L'Eau Bleue d'Issey Pour Homme'; 'Pour Homme Oud Noir'; 'Aqua Pour Homme'; 'Aramis for Men'; 'Terre d'Hermes EDT'; 'Tom Ford Noir Extreme'; 'Blue Label'; 'The One for Men EDT'; 'Valentino Uomo'; 'Dunhill Fresh'; 'Dior Homme Cologne 2013'; 'Joop! Homme'; 'L.12.12. Blanc'; 'A*Men Pure Havane'; 'Encre Noire A L'Extreme'; 'Grey Vetiver EDP'; 'Terre d'Hermes Parfum'; 'Bvlgari Man'; 'Tobacco Oud'; 'Bvlgari Man In Black'; 'Euphoria Intense'; 'Visit for Men'; 'Spicebomb Extreme'; 'Gucci by Gucci Pour Homme'; 'Roadster'; 'Bleu de Chanel EDP'; 'Nuit d'Issey Pour Homme'; 'Chrome Legend'; 'Acqua di Gio for Men'; 'Xeryus Rouge'; '212VIP for Men'; 'Mont Blanc Legend'; 'Chrome United'; 'He Wood'; 'Chic for Men'; 'Ombre Noire'; 'Luna Rossa'; 'Givenchy Gentleman'; 'Godolphin'; 'Dior Homme Sport'
C7	12	Women (100%)	Moderate (50%)	Floral (50%)	'Miracle'; 'Alien EDP'; 'Narciso Rodriguez for Her EDT'; 'Ange ou Demon (Etrange) Le Secret Elixir'; 'David Yurman Fragrance'; 'Gucci Flora EDP for women'; 'Cinema EDP'; 'Ange Ou Demon (Etrange) Le Secret'; 'Burberry London for Women'; 'Donna Trussardi 2011'; 'Versense'; 'Addict EDP 2014'
C8	11	Women (100%)	Warm (55%)	Oriental (55%)	'Si Eau de Parfum'; 'Libre'; 'La Panthere'; 'Jour d'Hermes'; 'Ange Ou Demon (Etrange) EDP'; 'Loverdose Red Kiss'; 'Twilly d'Hermes'; 'Jasmin Noir EDP'; 'Poison Girl'; 'Believe'; 'Angel Muse'

Table 6 (continued)

Community	Population (perfumes)	Dominant Gender (%)	Dominant Nature (%)	Dominant Fragrance group (%)	Members
C9	11	Men (100%)	Warm (73%)	Woody (46%)	'L'Homme Ideal EDP', 'Ultra Male', 'Dirty English', 'Overture Man', 'Prada L'Homme', 'La Nuit de l'Homme EDT', 'Valentino Uomo Intense', 'L'Homme Ideal', 'Boss Bottled', 'Prada L'Homme Intense', 'Acqua di Gio Profumo'
C10	15	Men (87%)	Warm–Moderate (Both 40%)	Woody (40%)	'John Varvatos Artisan', 'Erba Pura', 'Interlude for Men', 'Pardon', 'Allure Homme Edition Blanche EDP', 'Eclat d'Arpege Pour Homme', 'Leau Par Kenzo for Men', 'Sandstorm', 'Lyric for men', 'Dior Homme Intense', 'Declaration', 'Euphoria for Men', 'Antaeus', 'L'Homme Idéal Extrême', 'Gucci Guilty EDT Pour Homme'
C11	3	Men (100%)	Fresh - Warm - Moderate (Each 33.33%)	Woody (100%)	'Starwalker', 'Polo', 'Safari for Men'
C12	24	Women (100%)	Warm (42%)	Floral–Oriental (50–46%)	'Be Delicious for Women', 'Eclat d'Arpege for Women', 'Bright Crystal', 'Gucci Bloom', 'Chance EDP', 'Euphoria for Women', 'Lalique le Parfum', 'Idylle EDP', 'Halloween for Women', 'Manifesto EDP', 'Fantasy', 'La Petite Robe Noire EDP', 'Amethyst', 'Flowerbomb EDP', 'Beauty', 'La Vie Est Belle', 'Satine', 'La Nuit Tresor', 'Black Orchid', 'Pure Poison', 'L'Amour', 'Crystal Noir EDP', 'Good Girl', 'Tresor Midnight Rose'
C13	16	Men (63%)	Warm (50%)	Oriental (31%)	'Dior Homme Parfum', 'XJ 1861 Zefiro', 'Lalique White in Black', 'British Leather', 'Alexandria II', 'Nautica Voyage', 'CK One', 'L'Homme Lacoste', 'X for Men', 'Pour Un Homme', 'Tango', 'Dior Homme (2020)', 'Club De Nuit Sillage', 'H24', 'Kalan', 'Imitation For Man'
C14	10	Men (100%)	Warm (70%)	Oriental - Woody (Both 40%)	'Zino', 'Bentley Azure', 'Just Call Me Maxi', 'Pour Homme Intenso', 'Kouros', 'Halloween Man X', 'Encre Noire', 'Z-14', 'Pegasus', 'Emblem'
C15	5	Women (100%)	Warm (60%)	Oriental (80%)	'Hypnotic Poison', 'Hypnotic Poison EDP', 'Lilac Love', 'Black Opium', 'Honour for Women'
C16	5	Unisex (100%)	Warm (60%)	Oriental (60%)	'Sadonaso', 'Xerjoff 1888', 'Galloway', 'Isparzon', 'F**king Fabulous'
C17	3	Women (100%)	Moderate (67%)	Floral (67%)	'Coco Mademoiselle EDP', 'Bombshell', 'J'adore EDP'
C18	3	Women (100%)	Moderate (67%)	Oriental (67%)	'Elixir Des Merveilles EDP', 'Valentino Donna', 'Dune for Women'
C19	10	Women (90%)	Fresh–Warm (Both 40%)	Floral (70%)	'Viva la Juicy', 'Scandal', 'Eros Pour Femme EDP', 'Elle EDP', 'Burberry Body', 'Premier Jour', 'Bonbon', 'Jennifer Lopez Still', 'Mon Paris', 'Viva la Juicy Gold Couture'

Table 6 (continued)

Community	Population (perfumes)	Dominant Gender (%)	Dominant Nature (%)	Dominant Fragrance group (%)	Members
C20	7	Men (86%)	Fresh (57%)	Oriental - Woody - Citrus (Each 28.57%)	'Classic Black', 'Lalique White', 'Champion', 'Armani Code for Men', 'Eau Sauvage Parfum 2017', 'Versace Pour Homme', 'Bogart pour homme'

**Fig. 10** Distribution of fragrance groups across communities**Fig. 11** Distribution of perfume nature across communities**Fig. 12** Distribution of gender orientation across communities

Guerlain, and *Burberry Her*, complemented by niche offerings like *Gran Ballo* and *Lost Cherry*. Community 1 is very small (2 perfumes) but extremely homogeneous, grouping only youthful fresh florals (*Chance Eau Tendre*, *Daisy*) at 100% women, fresh, and floral. Community 2 brings together 6 masculine-oriented perfumes (83%), with fresh-oriental signatures such as *Aventus* and *Dunhill Desire*. Community 3 is among the largest clusters (27 perfumes), dominated by warm woody masculines (89% men), including icons like *Bleu de Chanel* and *Sauvage Elixir*, alongside niche examples like *Megamare*. In contrast, Community 4 is smaller (7 perfumes) but perfectly homogeneous in gender (100% unisex), composed mainly of warm orientals from niche houses such as *Amouage* (*Interlude 53*, *Jubilation*) and *Tuxedo*.

Community 5 is also one of the largest, with 74 perfumes and a majority unisex orientation (62%). Despite its size, it remains coherent through influential niche and hybrid favorites such as Amouage's *Reflection for Men*, Xerjoff's *Naxos*, and Maison Francis Kurkdjian's *Baccarat Rouge 540*, balanced by mainstream staples like *Dior Sauvage*. This blend suggests a bridge between designer and niche tastes. Community 6 is the largest cluster overall, (94 perfumes), almost entirely men's (87%), defined by warm woody structures. It combines heritage classics like *Fahrenheit* and *Azzaro Pour Homme* with modern powerhouses like *Tom Ford Noir Extreme* and *Spicebomb*, exemplifying the enduring masculine woody canon. Community 7 is small but highly coherent (12 perfumes, 100% women), balancing moderate floral signatures such as *Alien*, *Narciso Rodriguez for Her*, and *Burberry London for Women*. Community 8 (11 perfumes, 100% women) emphasizes warm orientals like *Si Eau de Parfum*, *Libre*, and *Poison Girl*, reflecting sensual statement-making femininity. Community 9 is compact (11 perfumes, 100% men), dominated by woody warm masculines (73%) including *La Nuit de l'Homme* and *Boss Bottled*, a cluster of modern urban sophistication. Community 10 (15 perfumes, 87% men) blends mainstream masculines like *Gucci Guilty* with niche entries like *Pardon*, unified by woody warm-moderate styles.

Community 11 is a tiny but highly consistent masculine trio (*Starwalker*, *Polo*, *Safari for Men*), all woody and versatile. Community 12 is a large feminine cluster (24 perfumes, 100% women), warm in nature and dominated by floral-orientals, with top sellers like *La Vie Est Belle*, *Flowerbomb*, and *Black Orchid*. Its size and high homogeneity confirm it as the mainstream warm feminine core. Community 13 (16 perfumes, 63% men) mixes mainstream and niche masculines, ranging from *CK One* to *Alexandria II*, linked by an oriental base that bridges consumer exploration across segments. Community 14 (10 perfumes, 100% men) unites bold warm woody-orientals such as *Kouros*, *Pegasus*, and *Encre Noire*, a cluster of assertive masculine archetypes. Community 15 is small but extremely homogeneous (5 perfumes, 100% women, 80% oriental), centered on sensual sweet orientals such as *Hypnotic Poison* and *Black Opium*.

Community 16 (5 perfumes, 100% unisex) reflects avant-garde exclusivity, with niche warm orientals like *Sadonaso* and *F**king Fabulous*, exemplifying experimental luxury. Community 17 is another small but very homogeneous group (3 perfumes, 100% women, moderate florals), including global bestsellers *Coco Mademoiselle* and *J'adore*, classic feminine elegance. Community 18 (3 perfumes, 100% women, moderate-oriental) contains *Elixir Des Merveilles* and *Valentino Donna*, refined, subtle feminines with less mainstream appeal. Community 19 (10 perfumes, 90% women) captures youthful gourmand-floral femininity, with playful scents like *Viva la Juicy* and *Scandal*, aligned with fashion-driven branding. Finally, Community 20 (7 perfumes, 86% men, mostly fresh) mixes citrus-woody masculines like *Armani Code* and *Versace Pour Homme*, versatile daily wear across age groups.

Taken together, these community-level interpretations reveal that the Louvain partitions are not arbitrary outcomes of modularity maximization but align closely with consumer-facing categories such as gender orientation, fragrance family, and market positioning. The presence of coherent clusters ranging from mainstream commercial cores (C12, C19) to niche exclusives (C4, C16) and heritage masculine archetypes (C6, C11, C14, C20) demonstrates that the detected communities reflect real-world perfume taxonomy and consumer segmentation. Importantly, the coherence of these

communities also shows that perfume recommendation cannot rely solely on declared fragrance notes or taxonomic families. Because our clusters are derived from positive user co-preferences, they capture deeper affinities that go beyond note lists, revealing structures that are more robust, interpretable, and consumer-relevant than what note-based similarity alone could provide. This extrinsic validation strengthens the robustness of our findings, highlighting the interpretive value of community detection in the perfume domain.

Conclusions and future work

In this study, we constructed a perfume co-preference network using user reviews from the Persian retail platform Atrafshan. By applying community detection algorithms, we uncovered meaningful clusters of similar perfumes, offering insights into user preferences in the fragrance industry. Our sentiment analysis approach, which integrated emojis and a user voting system, provided a more nuanced understanding of feedback.

We further refined the network by weighting edges based on average user ratings for Scent, Longevity, and Sillage. This refinement, combined with filtering for positive sentiment, significantly increased the modularity score across all community detection techniques and resulted in clearer and more meaningful perfume clusters. Beyond modularity, the interpretability analysis presented in Section 4, Results (Part C) demonstrated that these clusters are also interpretable, since they align closely with gender orientation, perfume nature, and fragrance families. This dual validation shows that the detected communities reflect both structural cohesion and consumer-facing categories.

A limitation of the present study is that we did not directly test whether sentiment-based homophily, understood as the tendency of perfumes to cluster together because of shared affective evaluations, explicitly drives the observed community structure. For example, we did not design a statistical test or null model that isolates sentiment effects from other structural factors such as co-review frequency. Instead, what we have shown is that incorporating sentiment signals, through textual reviews, emojis, and rating-based refinements, improves modularity scores and yields communities that are coherent and interpretable in terms of known fragrance taxonomies. The alignment we observed suggests that sentiment plays a structuring role in perfume networks, but a direct hypothesis test of sentiment-based homophily remains a compelling direction for future research.

Beyond this, several other future research directions can be pursued. One involves implementing a recommender system based on user profiles derived from the co-preference network. Another is incorporating real-time user interactions and developing a sentiment analysis model tailored specifically for perfume reviews to further enrich the results. Exploring external factors such as marketing campaigns or seasonal trends could provide valuable insights into the dynamic nature of consumer behavior. Finally, designing or adapting community detection criteria and modularity functions that explicitly account for sentiment information in their objective functions offers a promising path toward deeper integration of sentiment into network partitioning.

Overall, this research marks a significant step toward a data-driven understanding of preferences in the fragrance market and opens avenues for impactful applications in personalized retail experiences.

Author contributions

Kamand Kalashi conceived the study, developed the research idea, led the overall execution of the project, and took the lead in writing the manuscript. Sajjad Saed led the data collection and contributed equally with Kamand Kalashi to the methodology, coding, validation, formal analysis, and visualization. Babak Teimourpour supervised the project, provided guidance in methodology and validation, oversaw the interpretation of results, and offered constructive suggestions and feedback that guided the research direction. All authors read and approved the final manuscript.

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Data availability

The datasets analyzed in this study were scraped from the Atrafshan website, which is a Persian platform for perfume reviews and ratings. All data, including the User Reviews and Perfume Attributes Dataset, the Emoji Mapping Dataset, and the Sentiment Classification Results, are available in the *SentimentDrivenCommunityDetection* GitHub repository: <https://github.com/Kalashi-Saed-Collaborations/SentimentDrivenCommunityDetection>. The sentiment classification results are provided in three separate CSV files, each corresponding to a specific perfume attribute: Scent, Longevity, or Sillage. This dataset has been made publicly available to support reproducibility and further research.

Declarations

Competing interests

The authors declare that they have no competing interests.

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